A CASE-STUDY BASED REPORT

ON

Diabetes Risk Prediction Using Logistic Regression and Random Forest with Cross-Validation

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INDEX

Abstract

1. Introduction
   1.1 Background
   1.2 Motivation for the Project
   1.3 Overview of the Work Carried Out
   1.4 Scope and Responsible-Use Statement
2. Problem Statement
   2.1 Case-Study Scenario
   2.2 Central Case-Study Question
   2.3 Formal Problem Definition
   2.4 Objectives that Guide the Study
   2.5 Assumptions and Limitations
3. Dataset and Exploratory Data Analysis
   3.1 About the Dataset
   3.2 Purpose of EDA
   3.3 Preparation Steps
   3.4 EDA Exploration – Implementation
   3.5 Summary of EDA Findings
4. Algorithm Selection and Implementation
5. Experimental Results and Discussion
6. Conclusion and Future Scope

ABSTRACT

Diabetes is a chronic metabolic disease associated with elevated blood glucose levels. Delayed identification can increase the risk of complications such as cardiovascular disease, kidney failure, nerve damage and vision problems. This case study presents a machine learning approach for estimating diabetes risk from patient medical attributes. It compares Logistic Regression, used as an interpretable baseline classifier, with Random Forest, an ensemble classifier capable of learning non-linear relationships.

The project uses the PIMA Indians Diabetes Dataset, which contains attributes including pregnancies, glucose, blood pressure, skin thickness, insulin, body mass index (BMI), diabetes pedigree function and age. The workflow described in the project includes data preprocessing, handling zero values in selected medical measurements, feature scaling, model training and K-Fold Cross-Validation. The models are evaluated using accuracy, precision, recall, F1-score and ROC-AUC.

According to the experimental results presented in the project presentation, Logistic Regression achieved 77.2% accuracy, while the cross-validated Random Forest model achieved 88.4% accuracy. The presentation also reports higher precision, recall, F1-score and ROC-AUC for Random Forest. These results indicate that Random Forest performed better in the reported comparison. The system is intended as a decision-support and educational prediction prototype; it does not replace laboratory testing, clinical judgement or diagnosis by a qualified healthcare professional.

1. INTRODUCTION

1.1 Background

Diabetes is a chronic metabolic disorder in which the body cannot regulate blood glucose effectively. If it is not identified and managed in time, it may contribute to serious health problems, including cardiovascular disease, retinopathy, kidney complications and nerve damage. Conventional assessment uses clinical history, laboratory investigations and the judgement of healthcare professionals. Machine learning can complement this process by identifying patterns in existing patient data.

1.2 Motivation for the Project

The motivation is to explore how medical measurements can be used to estimate diabetes risk earlier and support preventive-care discussions. A data-driven model may help organize risk information and provide an additional reference for healthcare professionals. The project also provides a practical comparison between a linear classification method and an ensemble method.

1.3 Overview of the Work Carried Out

The work follows a machine learning pipeline: obtain the PIMA Indians Diabetes Dataset; inspect the attributes; prepare the data; train Logistic Regression and Random Forest classifiers; apply K-Fold Cross-Validation; and compare the models using standard classification metrics. The software environment specified in the project is Python 3.10 with Pandas, NumPy, Matplotlib, Seaborn and Scikit-learn. Visual Studio Code or Jupyter Notebook may be used as the development environment.

1.4 Scope and Responsible-Use Statement

The scope is limited to binary diabetes-risk prediction using the tabular features in the selected dataset and comparison of the two stated algorithms. Predictions should be treated only as screening support or an educational demonstration. Dataset patterns may not represent every population or clinical setting. A prediction must not be used on its own to start, stop or change treatment. Diagnosis and treatment decisions require appropriate medical evaluation.

2. PROBLEM STATEMENT

2.1 Case-Study Scenario

Consider a healthcare setting in which a large number of individuals provide routine measurements such as glucose level, blood pressure, BMI and age. Reviewing each record manually can take time, and relationships among several measurements may be difficult to identify consistently. The case study considers whether a machine learning model trained on historical patient records can classify a record into the dataset's diabetes outcome categories and help flag records that may need further clinical assessment.

2.2 Central Case-Study Question

Among Logistic Regression and Random Forest, which model provides stronger predictive performance for diabetes classification on the selected dataset when evaluated using K-Fold Cross-Validation and metrics such as accuracy, precision, recall, F1-score and ROC-AUC?

2.3 Formal Problem Definition

Let each patient record be represented by a feature vector containing the available medical attributes, and let the target variable represent the dataset's diabetes outcome class. The task is a supervised binary classification problem: learn a function that maps the input attributes to one of the two outcome classes. The learned models are compared on held-out or cross-validation predictions using the evaluation measures defined in the project.

2.4 Objectives that Guide the Study

- Study the medical attributes available in the PIMA Indians Diabetes Dataset.
- Inspect and prepare the dataset before model training.
- Handle invalid or zero-coded values in selected measurements as described in the project workflow.
- Apply feature scaling where required for the selected algorithms.
- Build Logistic Regression as a baseline classification model.
- Build a Random Forest classifier to learn more complex feature relationships.
- Use K-Fold Cross-Validation to assess model performance across multiple folds.
- Compare the models using accuracy, precision, recall, F1-score and ROC-AUC.

2.5 Assumptions and Limitations

The study assumes that the dataset's recorded attributes and outcome labels are suitable for training a supervised classifier. The report does not specify the exact fold count, random seed, hyperparameter values, train-test split ratio or sample-level clinical validation; these details are therefore not assumed here. The dataset is a limited tabular dataset and may not generalize to all age groups, ethnic groups, locations or healthcare systems. The reported metrics describe the project experiment and should not be interpreted as guaranteed real-world clinical accuracy.

3. DATASET AND EXPLORATORY DATA ANALYSIS

3.1 About the Dataset

The project identifies the PIMA Indians Diabetes Dataset, obtained from a public Kaggle/UCI repository, as its data source. The dataset contains patient-related numerical attributes used to predict the diabetes outcome. The features listed in the project are:

- Pregnancies – number of pregnancies recorded.
- Glucose – glucose measurement.
- Blood Pressure – diastolic blood pressure measurement.
- Skin Thickness – skin-fold thickness measurement.
- Insulin – insulin measurement.
- BMI – body mass index.
- Diabetes Pedigree Function – a dataset feature representing hereditary diabetes-related information.
- Age – age of the individual.

The target is the diabetes outcome class in the dataset. The supplied project documents do not state the final number of records used after preprocessing; this report therefore does not add a record count.

3.2 Purpose of EDA

Exploratory Data Analysis (EDA) is used to understand the dataset before training. It helps inspect the range and distribution of each feature, identify missing or unusual values, observe possible relationships between medical attributes and the target, and decide which preparation steps are needed. EDA also helps detect data-quality issues that could affect model performance.

3.3 Preparation Steps Actually Described in the Project

1. Inspect the dataset and its medical feature columns.
2. Treat zero values in Glucose, Insulin and BMI as invalid or missing measurements where appropriate.
3. Use median imputation for the selected zero-coded/missing measurements, as specified in the presentation methodology.
4. Apply StandardScaler feature scaling where required.

5. Prepare the features and target for supervised classification.

3.4 EDA Exploration – Implementation

The project lists Pandas and NumPy for data handling and Matplotlib and Seaborn for visualization. A suitable EDA implementation for this workflow includes checking dataset dimensions and data types, viewing descriptive statistics, counting missing or zero-coded measurements, plotting feature distributions, and examining feature relationships. The supplied presentation specifies the preprocessing approach but does not provide individual EDA plots or numerical EDA findings; no such values are invented in this report.

3.5 Summary of EDA Findings

The project identifies Glucose, BMI and Age as important predictors in its reported findings. The presentation also emphasizes that zero values in selected medical measurements require preprocessing. These observations motivate careful data cleaning and the comparison of models that can represent different types of relationships in the data.

4. ALGORITHM SELECTION AND IMPLEMENTATION

4.1 Logistic Regression

Logistic Regression is used as the baseline binary classifier. It estimates the probability of an outcome class through a logistic function applied to a weighted combination of input features. It is relatively simple and interpretable, making it useful for understanding the contribution of input variables. Its limitation is that a basic Logistic Regression model uses a linear decision boundary and may not capture complex non-linear interactions between medical attributes.

4.2 Random Forest

Random Forest is an ensemble learning algorithm that builds multiple decision trees and combines their predictions, commonly through majority voting for classification. It can represent non-linear relationships and interactions among features. Combining multiple trees can reduce the variance associated with relying on a single decision tree. Its performance still depends on appropriate data preparation and model settings.

4.3 K-Fold Cross-Validation

K-Fold Cross-Validation divides the available training data into K parts, or folds. In each iteration, one fold is used for validation and the remaining folds are used for training. The process repeats until each fold has served as the validation fold. This provides a more robust assessment than relying on only one split. The presentation mentions 5-Fold / 10-Fold validation as options; it does not identify which fold count was used for the displayed final results.

4.4 Implementation Workflow

6. Load the PIMA Indians Diabetes Dataset using Pandas.
7. Inspect columns, data types and data quality.
8. Handle selected zero-coded measurements using median imputation.
9. Scale features using StandardScaler where required.
10. Separate input features from the diabetes outcome target.
11. Train Logistic Regression and Random Forest classifiers.
12. Evaluate using K-Fold Cross-Validation and the selected classification metrics.
13. Compare the reported model metrics and identify the stronger-performing model.

4.5 Software and Tools

Component	Requirement / Tool
Operating System	Windows 10 / 11
Programming Language	Python 3.10
IDE	Jupyter Notebook / Visual Studio Code / PyCharm
Data Processing	Pandas, NumPy
Visualization	Matplotlib, Seaborn
Machine Learning	Scikit-learn
Model Persistence	Joblib / Pickle (listed in presentation)
Version Control	Git & GitHub

5. EXPERIMENTAL RESULTS AND DISCUSSION

The following values are transcribed from the Experimental Results & Comparison slide in the supplied presentation. They are presented as the project's reported results, not as independently recomputed measurements.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	77.2%	73.5%	68.1%	70.7%	0.81
Random Forest (Cross-Validated)	88.4%	86.2%	83.5%	84.8%	0.92

5.1 Interpretation

In the reported comparison, Random Forest achieved higher values than Logistic Regression for all five listed evaluation measures. Its accuracy was 88.4%, compared with 77.2% for Logistic Regression. The presentation attributes the stronger performance to Random Forest's ability to capture non-linear feature interactions and reduce overfitting through an ensemble of decision trees. The reported ROC-AUC values are 0.92 for Random Forest and 0.81 for Logistic Regression.

The presentation identifies Glucose, BMI and Age as the strongest predictors in the project. This is a project-reported finding; the supplied materials do not include a feature-importance chart or detailed numerical importance values.

5.2 Result Considerations

The metrics should be interpreted in the context of the selected dataset and the evaluation procedure. For a reproducible experiment, the implementation should record the exact fold count, random seed, preprocessing pipeline, classifier settings and whether preprocessing was fitted inside each cross-validation fold. Those implementation details are not fully specified in the provided documents.

6. CONCLUSION AND FUTURE SCOPE

This case study demonstrates a machine learning workflow for diabetes-risk classification using the PIMA Indians Diabetes Dataset. Logistic Regression provides an interpretable baseline, while Random Forest provides an ensemble approach for learning more complex patterns. Based on the results reported in the project presentation, Random Forest outperformed Logistic Regression across accuracy, precision, recall, F1-score and ROC-AUC. K-Fold Cross-Validation is included to improve the reliability of model evaluation.

The project illustrates how predictive analytics may support early risk identification and preventive-care awareness. However, it is a prototype based on a particular public dataset and must not be treated as a diagnostic system. Further work may include validating the model on independent and more diverse datasets, documenting reproducible training settings, tuning model hyperparameters, evaluating calibration and subgroup performance, and developing a secure user interface for clinician-supervised use.

Signature of the Team Members

Signature of the Guide

Course Instructor